

CareerPilot AI

Professional Software Engineering Handbook

This handbook is intended to document the complete lifecycle of designing, developing, testing, deploying and scaling CareerPilot AI using modern engineering practices.

Handbook Structure

Part	Contents
I	Requirements, Vision, SRS, User Stories
II	Architecture, C4 Diagrams, ER Diagrams, API Standards
III	Spring Boot Backend
IV	Angular Frontend
V	Database Design & Flyway
VI	Testing (JUnit, Mockito, REST Assured, Selenium, Cucumber)
VII	Docker, CI/CD, GitHub Actions
VIII	Cloud Deployment & Kubernetes
IX	Monitoring, Logging, Scaling, Security
X	Resume, GitHub Documentation & Interview Preparation

Engineering Standards

- Clean Architecture and SOLID principles
- Conventional commits and GitFlow
- Test pyramid (Unit → Integration → API → UI)
- CI/CD on every merge
- Docker-first deployment, Kubernetes later
- API-first development with OpenAPI
- Secure coding (JWT, validation, RBAC, secrets)
- Observability (logs, metrics, health checks)